# Skills Development for Adolescent Girls

Investment case, East Java, Indonesia

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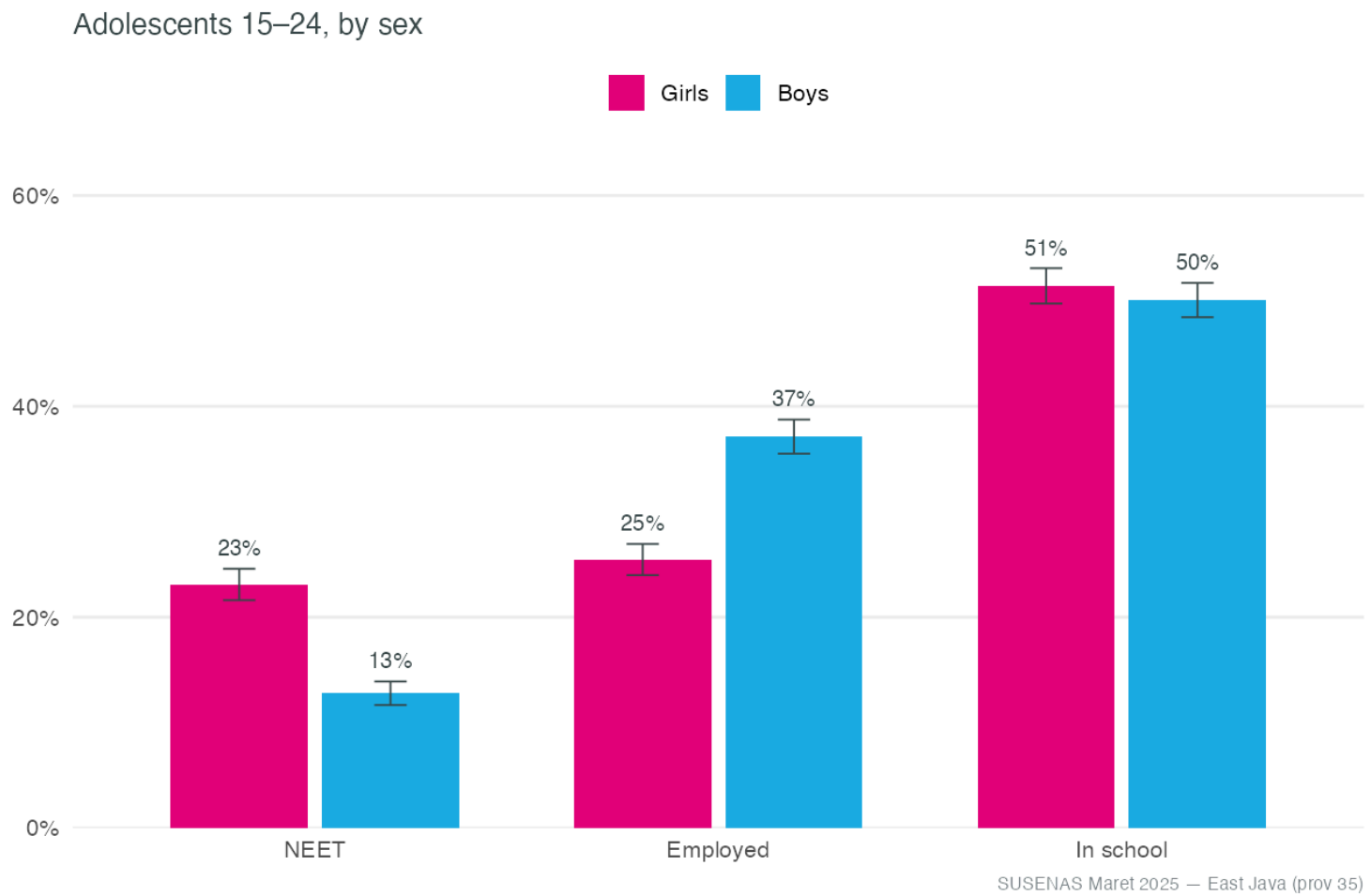
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# 1 · The problem

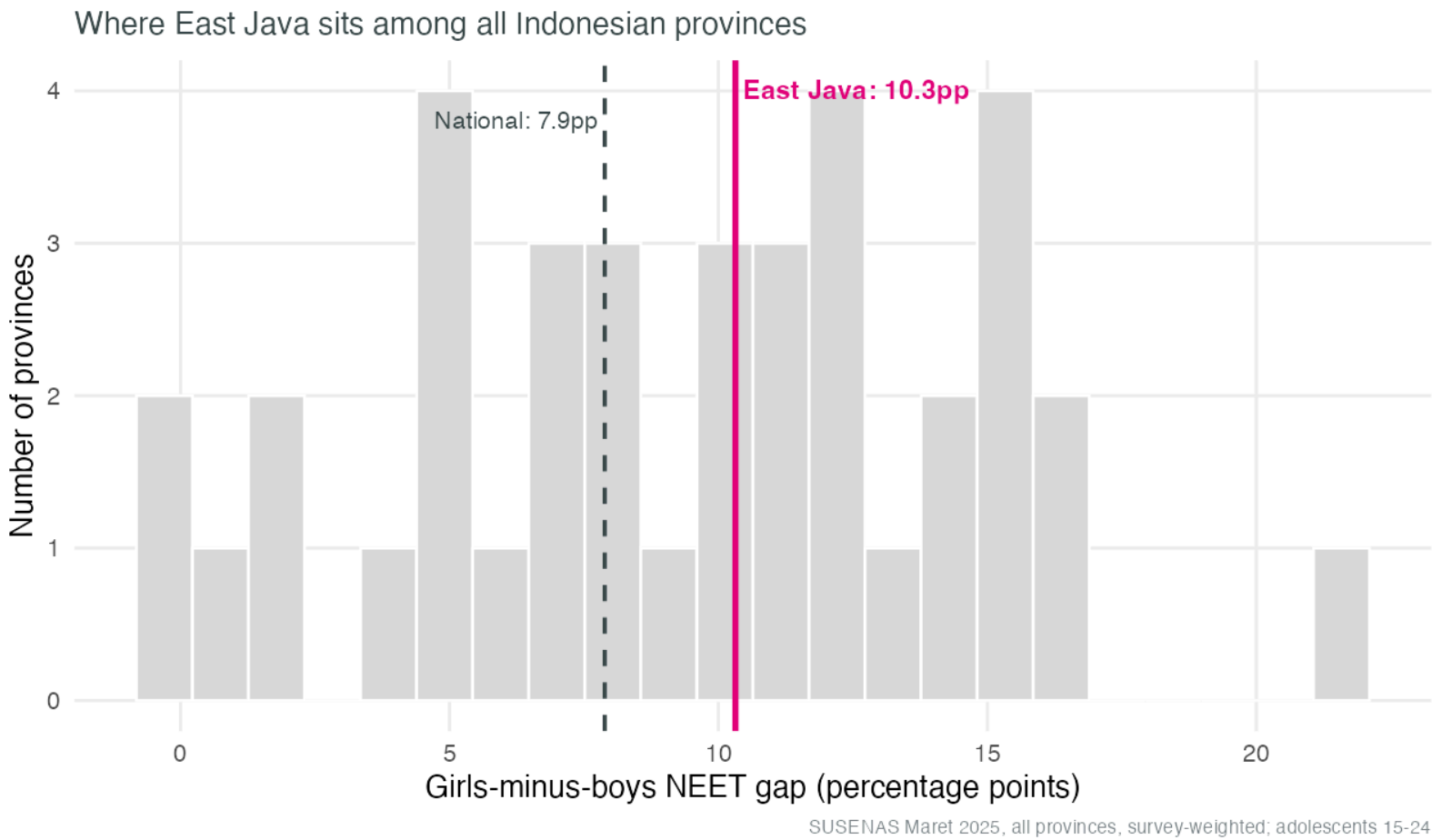
# Girls have the same school attendance rate as boys, but their NEET rate is close to double




*Data source: SUSENAS Maret 2025, East Java*

**Notes:** NEET: not in employment, education or training. Girls 15–24: 51% in school (boys 50%), 23% NEET (boys 13%), 25% employed (boys 37%). The gap opens at the transition out of school. SUSENAS Maret 2025, survey-weighted, 95% CIs.

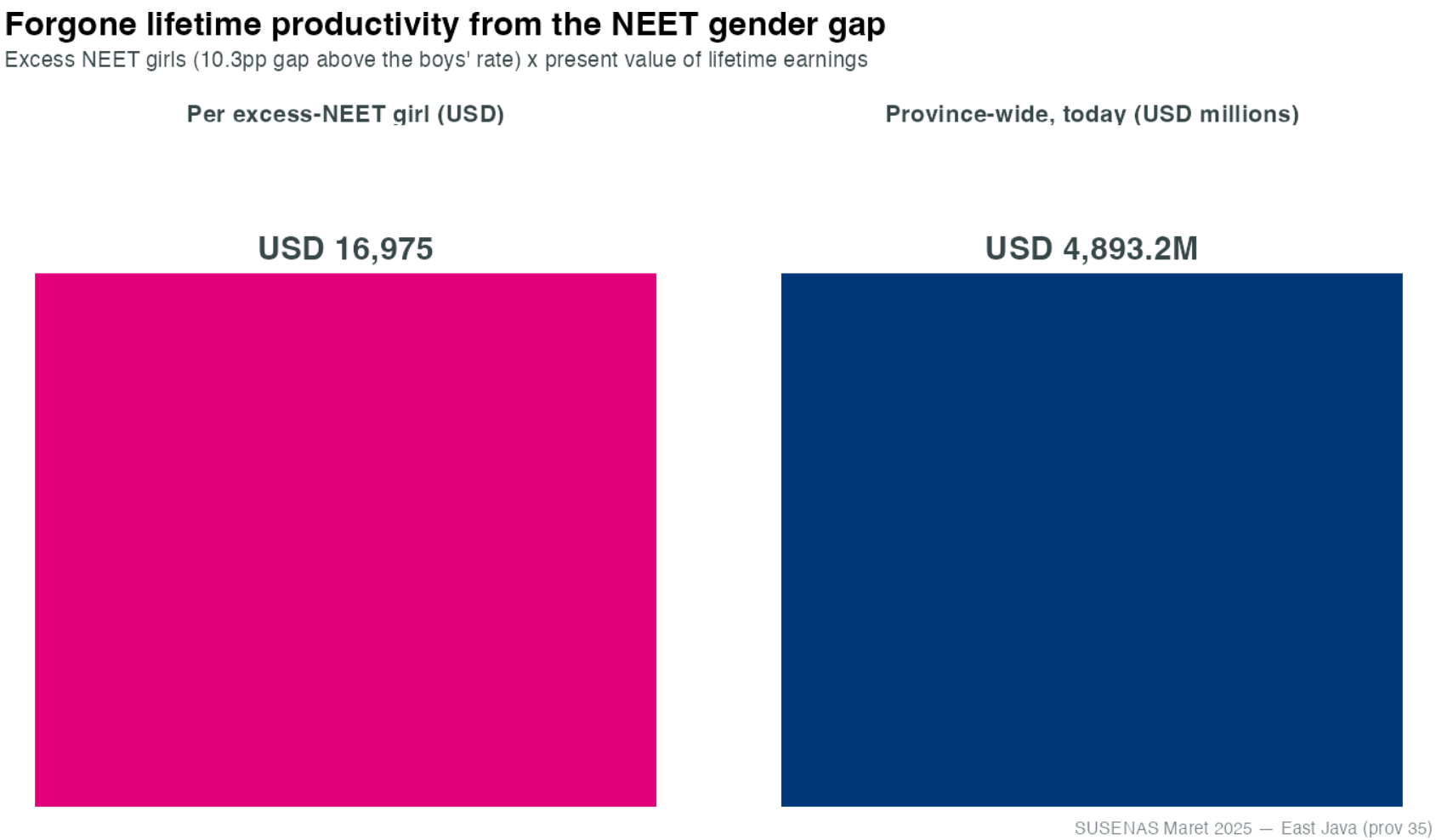
# East Java’s NEET gender gap is wider than the national average




*Data source: SUSENAS Maret 2025, all provinces*

**Notes:** Girls-minus-boys NEET gap: 10.3 points in East Java against 7.9 points nationally. Same survey, same definition, all provinces.

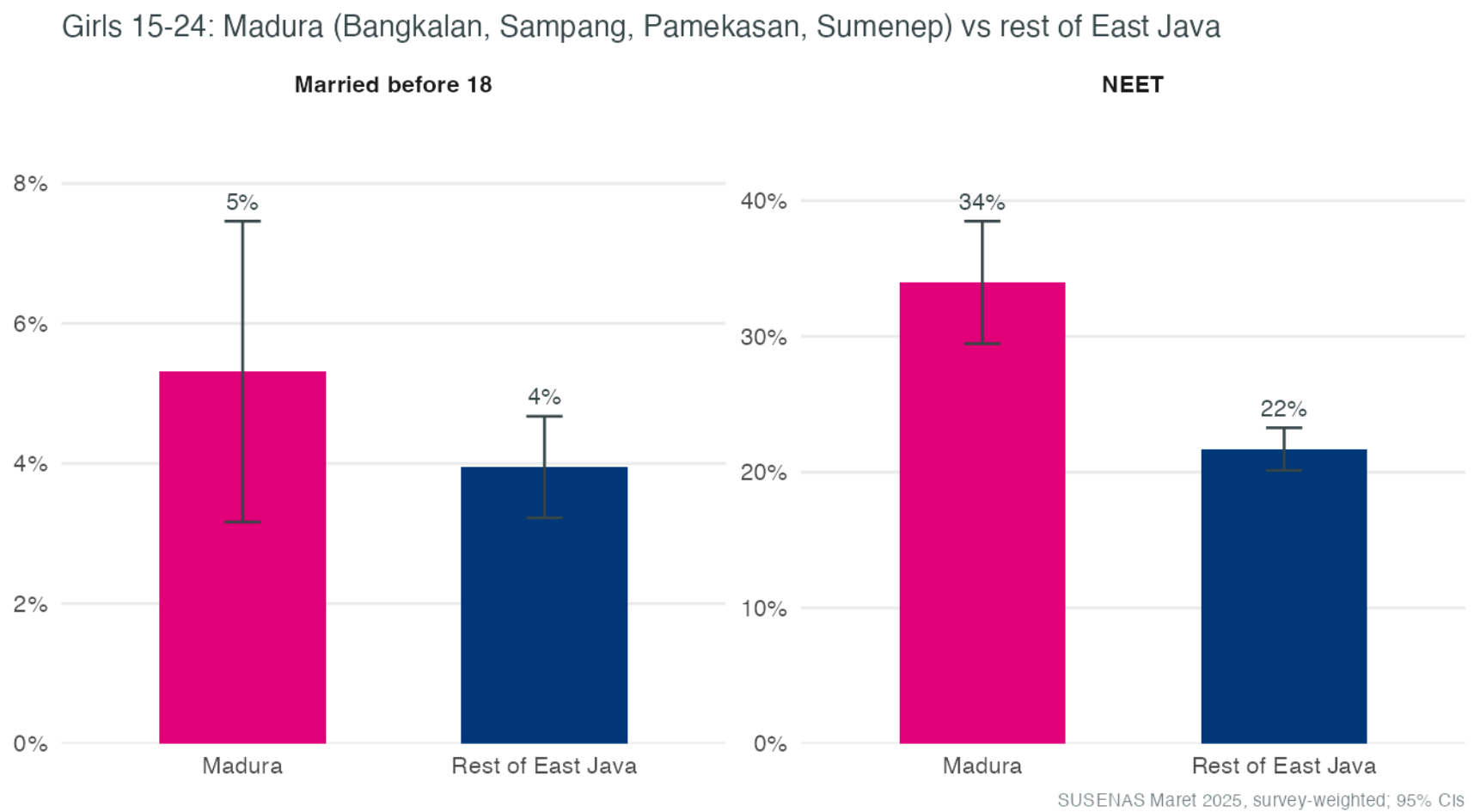
# The gap costs an estimated USD 4.9 billion in forgone lifetime productivity




*Data source: SUSENAS Maret 2025, East Java; earnings assumptions as in section 3*

**Interpretation:** About 288,000 girls are NEET in excess of the boys' rate. Each would be expected to earn about USD 17,000 more over a working lifetime (present value) if she had the boys' NEET rate.

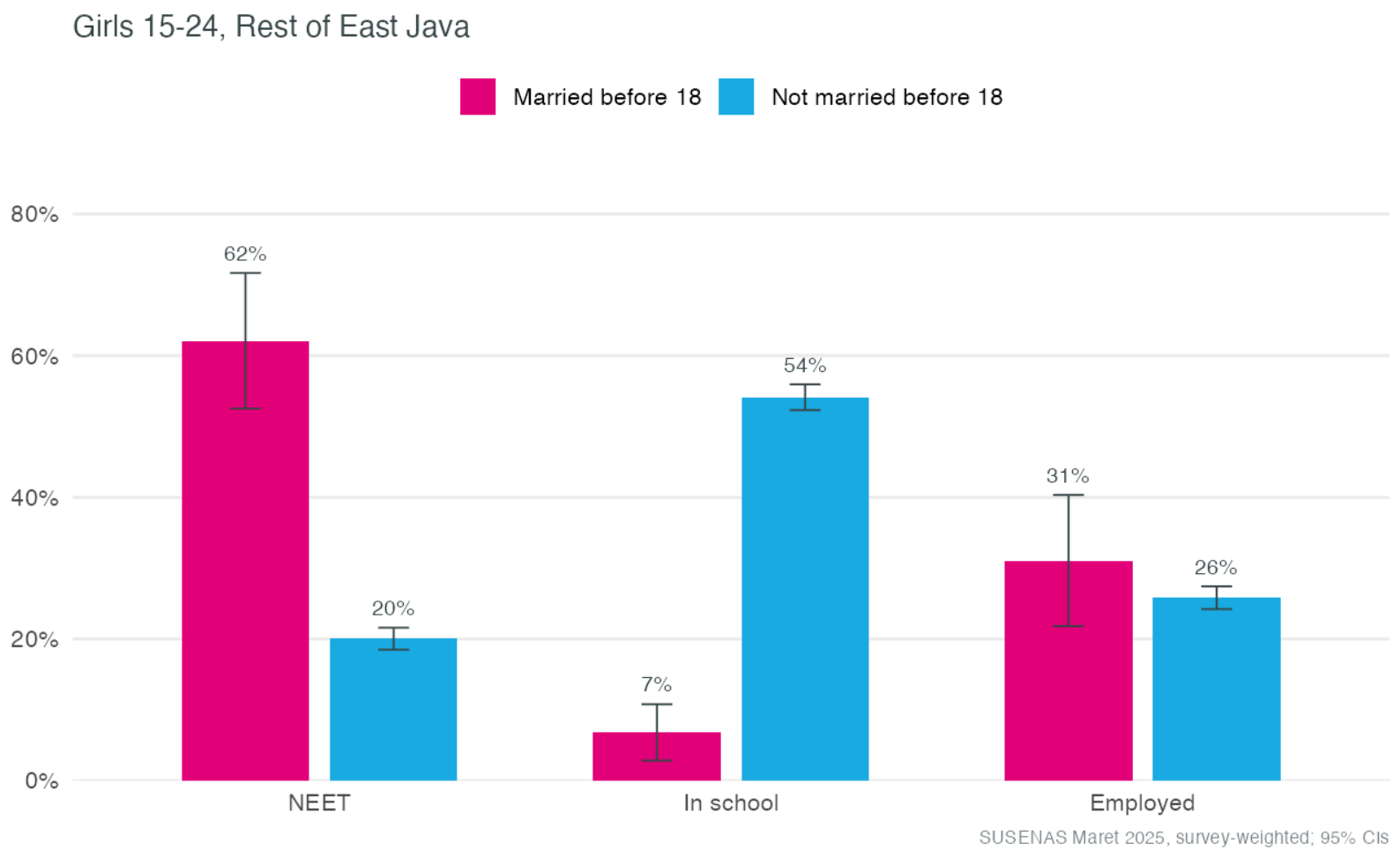
# Madura has higher NEET for girls and boys, with a similar gender gap




*Data source: SUSENAS Maret 2025, East Java*

**Interpretation:** Girls' NEET is 34% in Madura (Bangkalan, Sampang, Pamekasan, Sumenep) against 22% in the rest of East Java; boys' NEET is 23% against 12%. The gender gap is similar (11 vs 10 points), and fewer girls are in school (45% vs 52%). Madura is a priority area for a pilot.

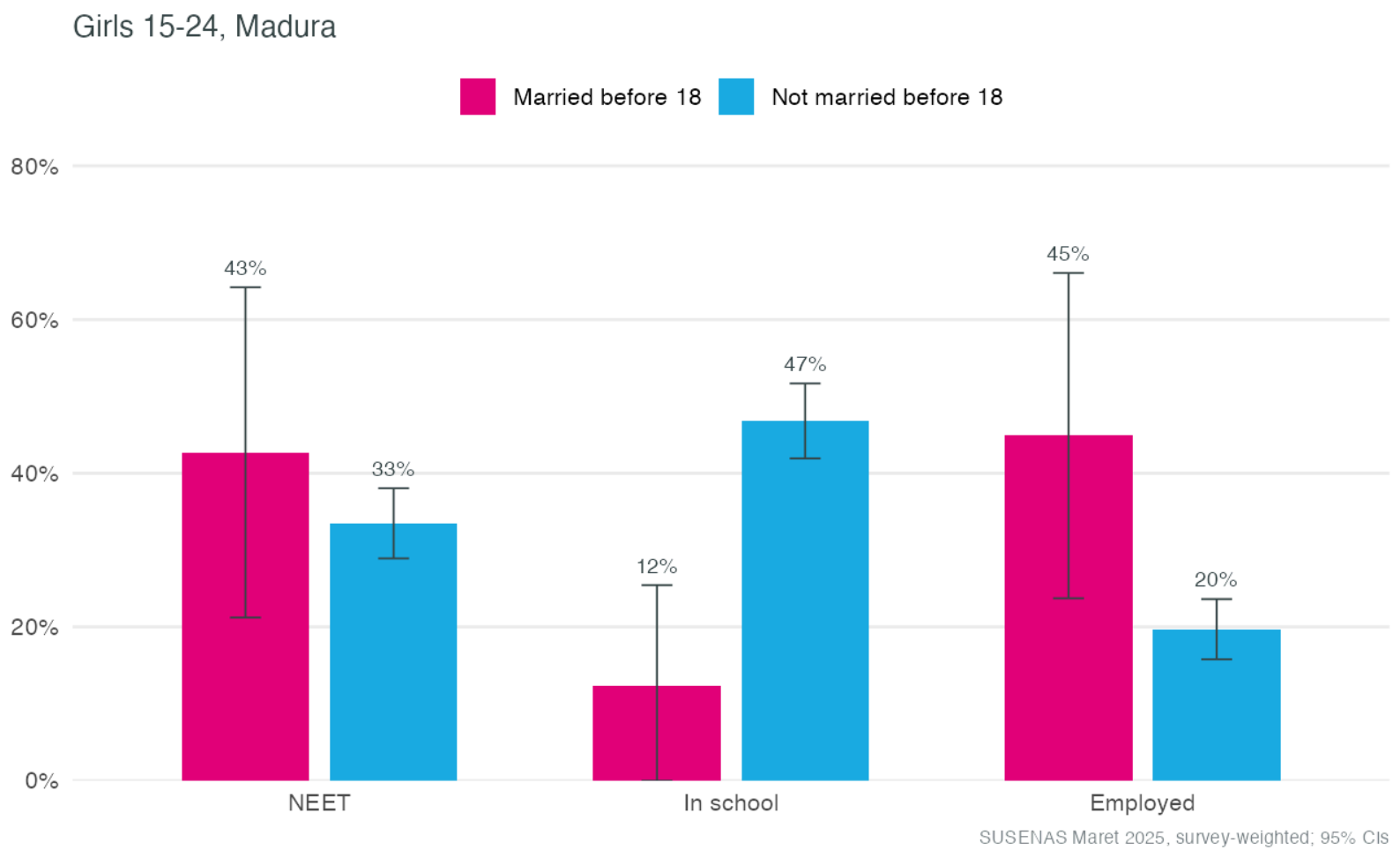
# Child marriage and outcomes: rest of East Java




*Data source: SUSENAS Maret 2025, East Java*

**Notes:** Girls 15–24 outside Madura, by whether they married before 18, survey-weighted (95% CIs). Regression tests by region are in the appendix. Early pregnancy before 18 overlaps almost entirely with child marriage in this data, so only child marriage is shown. Association, not causation.

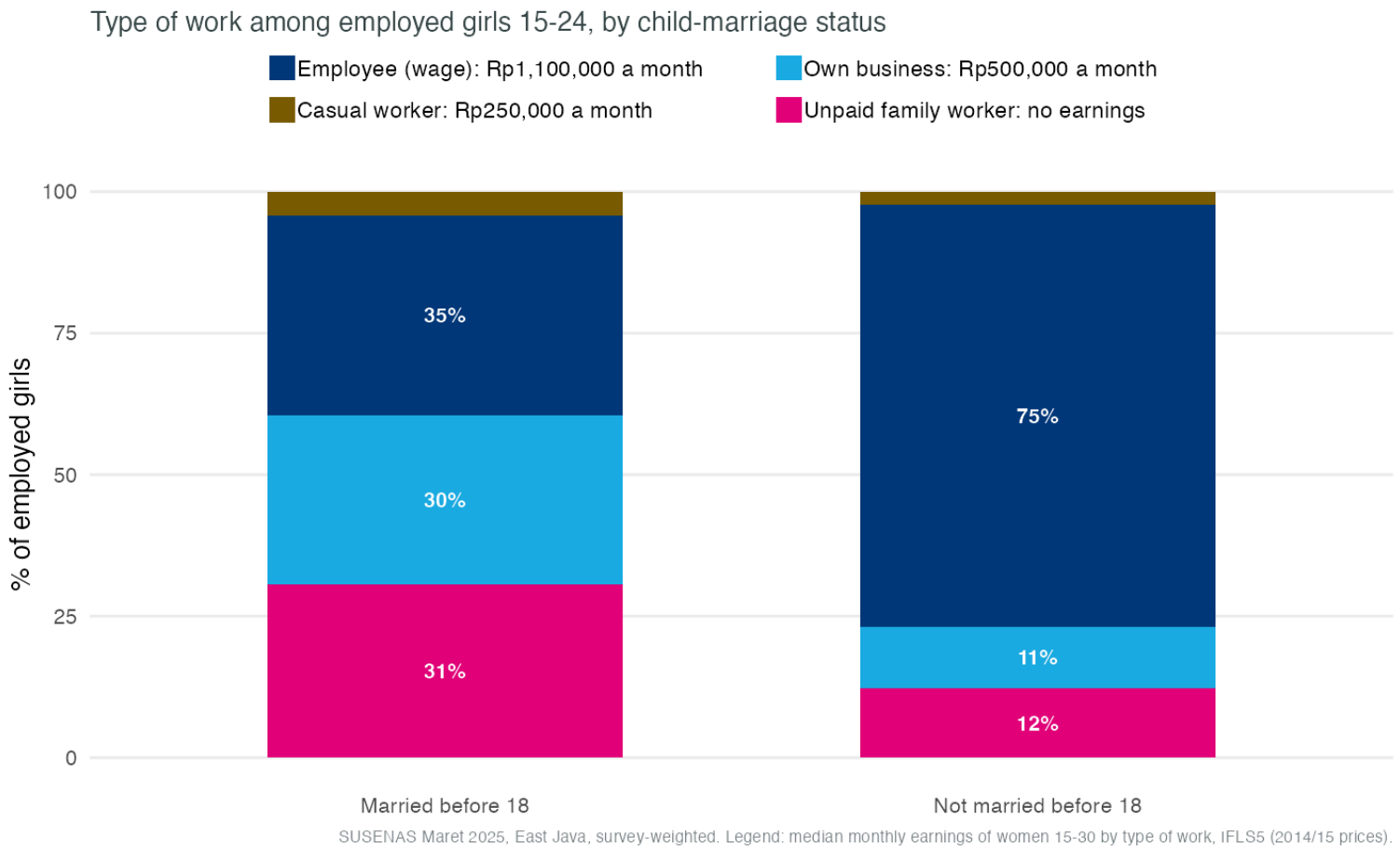
# Child marriage and outcomes: Madura




*Data source: SUSENAS Maret 2025, East Java*

**Notes:** Girls 15–24 in Bangkalan, Sampang, Pamekasan and Sumenep. Fewer girls, so wider intervals. Association, not causation.

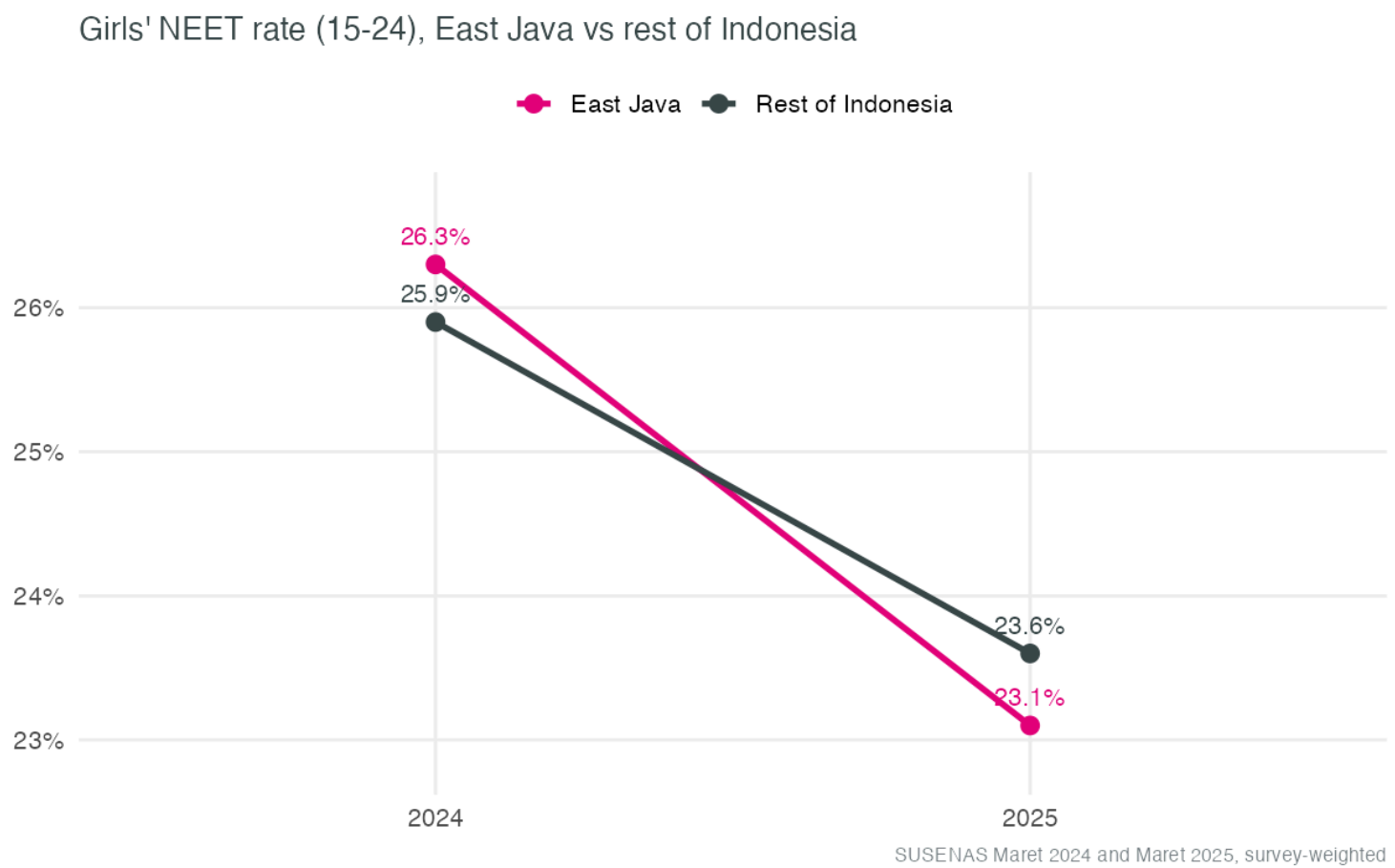
# Married girls who work do more unpaid family work and less wage employment




*Data source: SUSENAS Maret 2025 (type of work); IFLS5 2014/15 (earnings)*

**Interpretation:** Among employed girls married before 18, 35% are wage employees (others 75%) and 31% are unpaid family workers (others 12%); 22% work under 20 hours a week (others 12%). Based on 78 employed married girls, so indicative. SUSENAS has no earnings; the appendix shows earnings by type of work for young women from IFLS5.

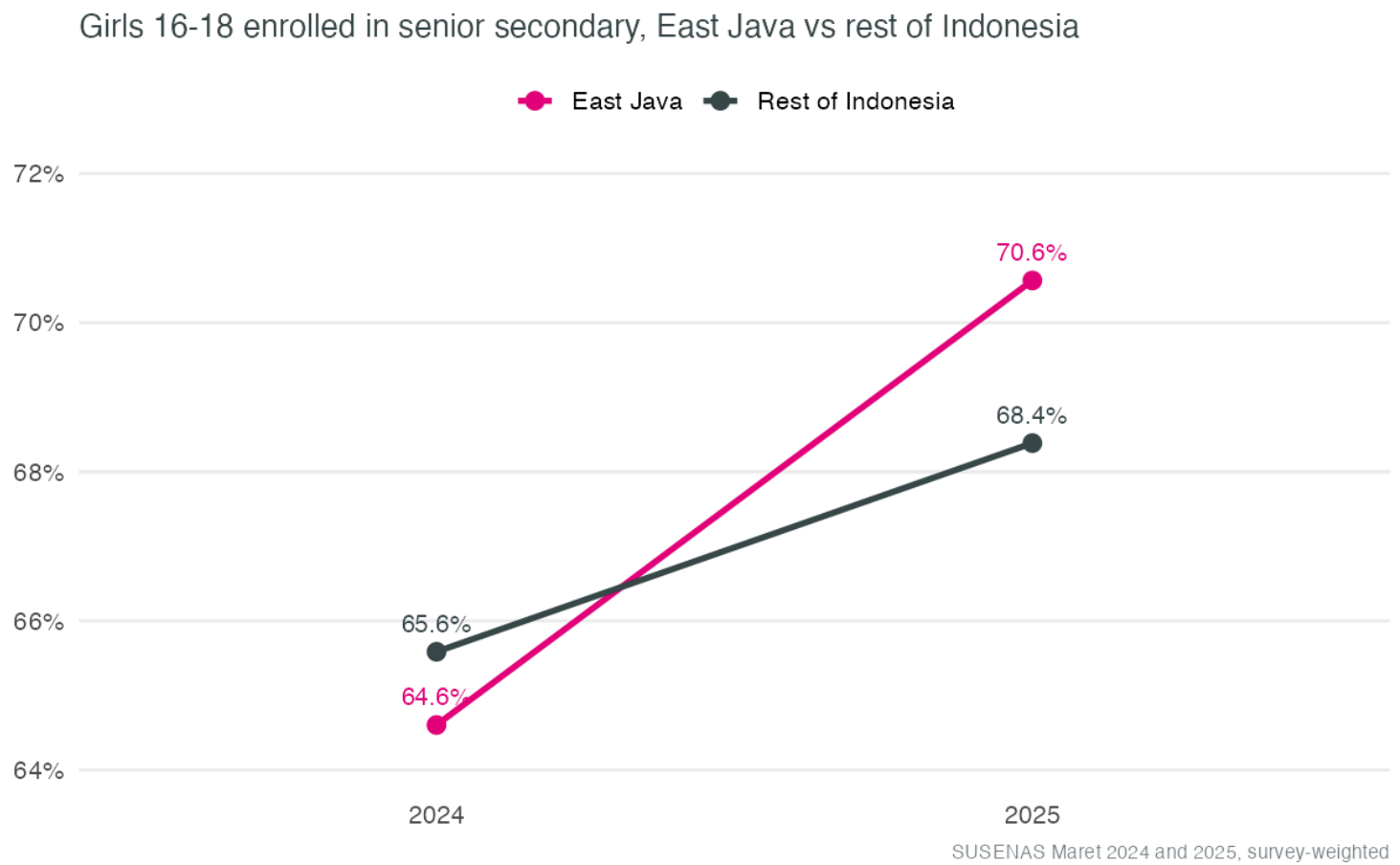
# Girls’ NEET in East Java and the rest of Indonesia, 2024 and 2025




*Data source: SUSENAS Maret 2024 and 2025, all provinces*

**Interpretation:** Girls’ NEET (15–24) fell 3.2 points in East Java and fell 2.3 points in the rest of Indonesia; child marriage fell 1.3 points and fell 0.7 points; tertiary enrolment at 18–20 rose 3.2 points and rose 5.0 points. Both years use the same definition (any work in the last week, or temporarily absent from a job; not attending school). Survey trends cannot separate a programme from wider changes; a comparison of programme and non-programme districts is pending.

# Senior-secondary enrolment rose faster in East Java than in the rest of Indonesia




*Data source: SUSENAS Maret 2024 and 2025, all provinces*

**Interpretation:** Share of 16–18 year olds enrolled in senior secondary rose 6.0 points for girls in East Java against 2.8 in the rest of Indonesia (boys: 5.8 against 2.6). The faster rise applies to both sexes, so it points to something affecting East Java's schools generally rather than a girls-specific programme. The district comparison can test whether programme districts drove it.

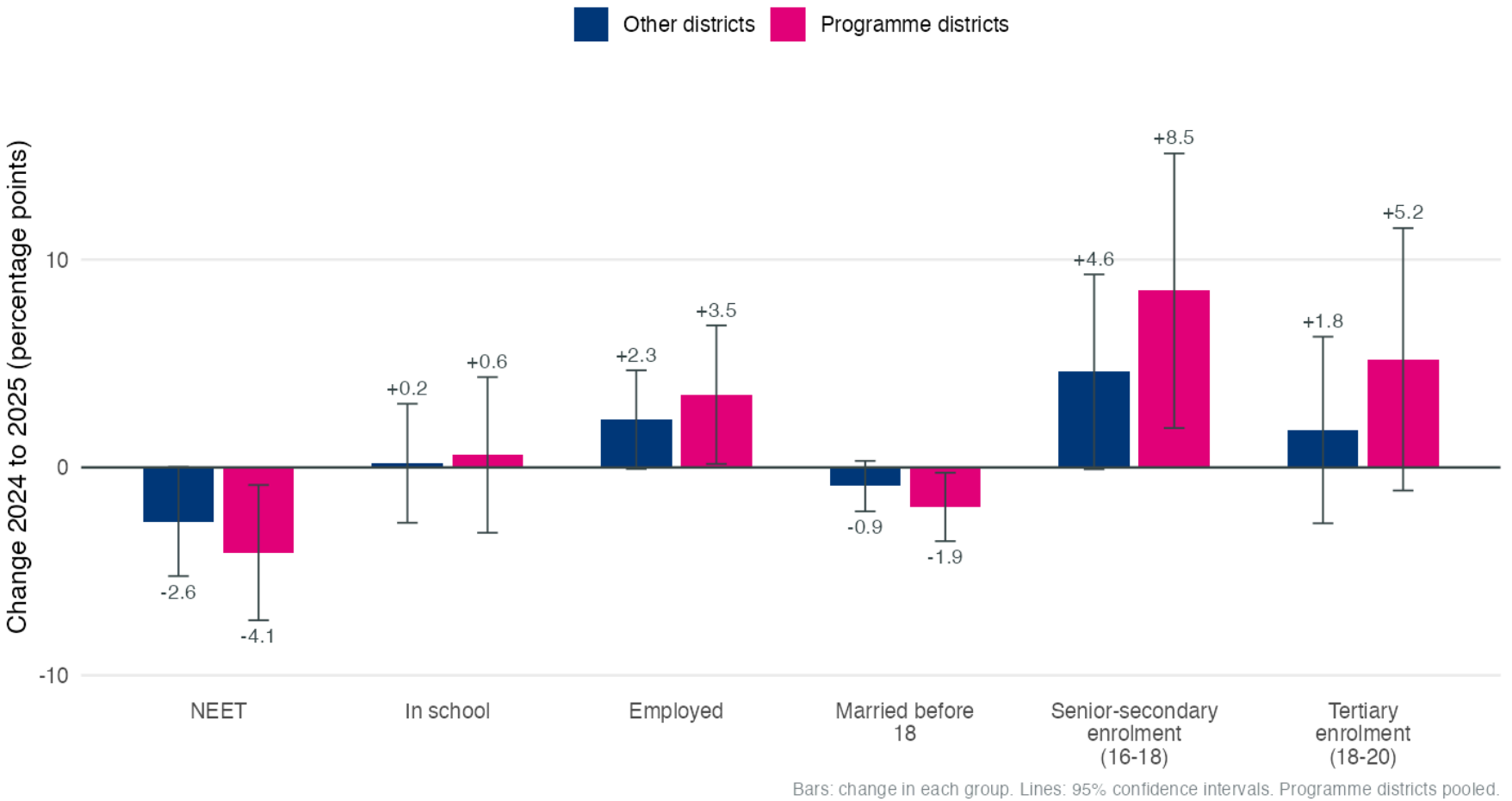
# What changed for girls between 2024 and 2025

| Indicator | East Java 2024 | East Java 2025 | East Java change | Rest of Indonesia change |
|---|---|---|---|---|
| NEET, 15-17 | 9.5 | 6.8 | -2.8 | -1.3 |
| NEET, 18-21 | 28.6 | 24.7 | -3.9 | -3 |
| NEET, 22-24 | 40.7 | 37.5 | -3.2 | -2.2 |
| NEET, 22-24 with tertiary certificate | 32.7 | 34.3 | 1.6 | -1.1 |
| NEET, 22-24 without tertiary certificate | 42.7 | 38.5 | -4.3 | -2.5 |
| In school, 15-24 | 51.1 | 51.4 | 0.4 | 0.5 |
| Employed (not in school), 15-24 | 22.7 | 25.5 | 2.8 | 1.7 |
| Own business, % of employed | 12.6 | 11.8 | -0.8 | -0.2 |
| Employee, % of employed | 71.6 | 72.5 | 0.9 | -0.1 |
| Unpaid family worker, % of employed | 13.3 | 13.3 | 0 | 0.6 |
| Trade or transport sector, % of employed | 27.9 | 32.4 | 4.5 | -0.5 |

*Data source: SUSENAS Maret 2024 and 2025, all provinces*

**Interpretation:** Girls 15–24, same definitions in both years. The fall came mainly from more girls working (+2.8 points), not more girls in school (+0.4). NEET rose slightly among 22–24 year olds with a tertiary certificate, so the national internship programme explains little of the change in East Java. The share of working girls in trade or transport rose 4.5 points, against no change elsewhere, consistent with growth in informal and platform-based work.

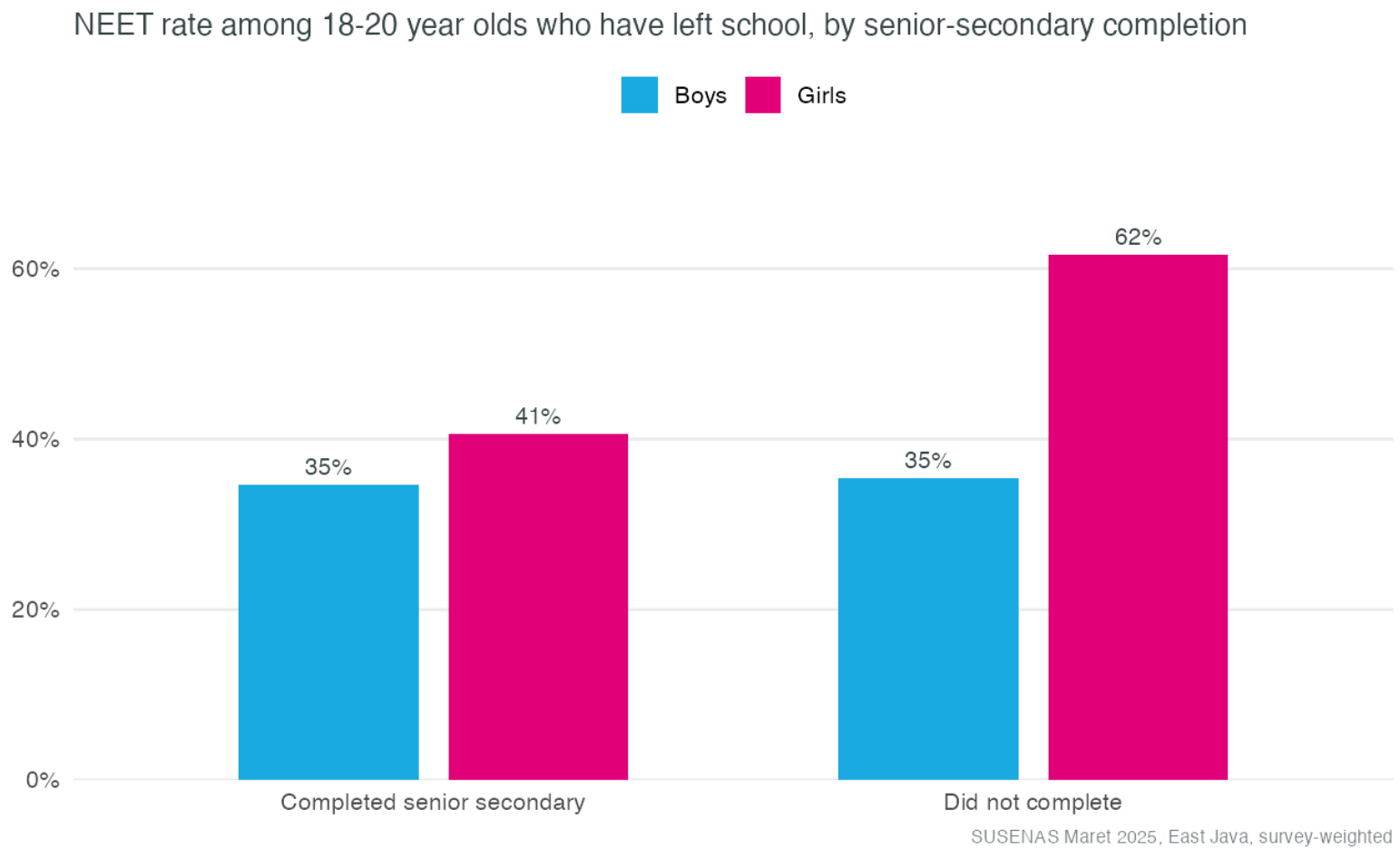
# Programme districts compared with the rest of East Java




*Data source: SUSENAS Maret 2024 and 2025, East Java; programme districts from UNICEF Indonesia*

**Interpretation:** Twelve programme districts pooled, against the other 26. Every indicator improved more in programme districts, but none of the differences is statistically significant: for girls' NEET the difference is -1.5 points (95% confidence interval -5.7 to 2.7, p = 0.487). Lines on the bars show 95% confidence intervals. Programme districts were selected rather than assigned at random, and if the programme started before March 2024 part of its effect is already in the baseline, so this is a descriptive comparison, not an impact estimate.

# Completing senior secondary is associated with 20 points lower NEET for girls




*Data source: SUSENAS Maret 2025, East Java*

**Interpretation:** Among 18–20 year olds who have left school, girls who completed senior secondary have a NEET rate of 41%, against 62% for those who did not. Adjusted for age and area (girls 18–24), completion is associated with a 20-point lower NEET rate ($p<0.001$). For boys there is no observed difference. Keeping girls in school to completion is the programme's main lever.

# 2 · Theory of change

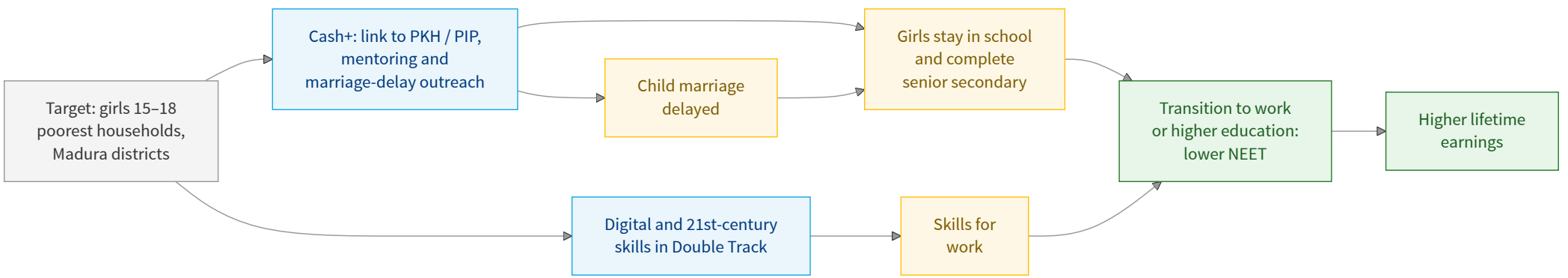
# Evidence from Indonesia

| Programme | Design | Result | Relevance |
|---|---|---|---|
| **PKH** cash transfer (Priebe & Sumarto, 2025) | Quasi-experimental | Child marriage −3.5pp among girls | A cash component reduces child marriage |
| **PIP** education assistance (Ulfa & Rezki, 2024) | Matching and IV | School participation +11.4pp (pooled, not girls-specific) | Anchors the effect used in the ROI, with a transferability assumption |
| **PIP**, girls' senior secondary (Caniago et al., 2021) | Observational regression | Girls' senior-secondary enrolment +24.6pp (adjusted association) | Targets the stage where girls drop out |
| **Vocational vs general secondary schooling** (Pritadrajati, 2022) | Quasi-experimental | Female LFPR +3.4pp (significant at 10%); income +5.6% (not significant) | Used only as an uncertain scenario in the ROI |

*Data source: published evaluations (see references)*

**Interpretation:** In SUSENAS 2025, girls in PKH households have lower child-marriage rates than other girls (3% vs 4–6%), even though PKH targets poorer households. Consistent with the PKH evidence, though not a causal estimate.

# Theory of change




*Data source: published evaluations and this analysis*

- Cash transfers reduce girls' child marriage (PKH, Indonesia).
- Education assistance raises school participation and girls' senior-secondary enrolment (PIP, Indonesia).
- Completing senior secondary is associated with 20 points lower NEET for girls (this analysis).
- Skills to earnings is the weakest link: the Indonesian estimate is not statistically significant.
- Mentoring and outreach: international evidence for girls only.
- Cash+ links girls to existing PKH and PIP; no new transfer.

# 3 · Projected returns (ex-ante)

# Key assumptions

| Assumption | Value | Source |
|---|---|---|
| Reduction in girls' NEET | 7.4 points | PIP school-participation effect (+11.4pp, Ulfa & Rezki 2024) × 65% of girls who leave school before completing senior secondary being NEET (SUSENAS 2025) |
| Extra schooling for a girl kept in school | 3 years | Senior-secondary cycle |
| Earnings gain per year of schooling | 8.0% | IFLS5 wage return for women, controlling for region, parents' education and selection into paid work |
| Female earnings | Rp20 million/year | Below East Java minimum wage 2025, reflecting informal work |
| Probability of earning (employment rate, women 25–54, East Java) | 62% | SUSENAS Maret 2025 |
| Earnings growth / discount rate | 2% / 3% per year | Career growth; 3% real, with 2% and 5% in sensitivity |
| Skills earnings premium (scenario only, not in the central case) | 5.6% for 10 years | Point estimate for vocational vs general schooling (Pritadrajati 2022), not statistically significant |
| Programme cost | USD 200 per girl | USD 58 per year for 3 years, plus 15% overhead; Cash+ links to existing PKH/PIP |

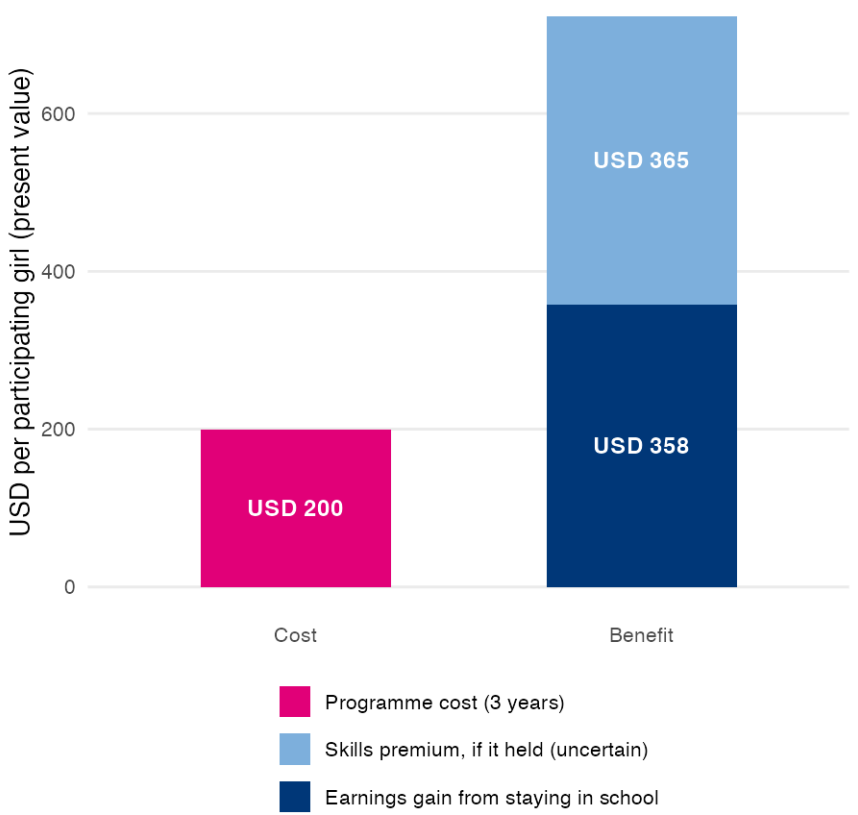
*Data source: modelled projection (ex-ante)*

# How the return is built

| Step | Value |
|---|---|
| Extra years of schooling for a girl kept in school | 3 |
| Earnings gain per year of schooling | 8.0% |
| Earnings gain for that girl | 24% |
| Female earnings (USD/yr) x labour force participation | 1,266 x 62% |
| Extra earnings per year (USD) | 188 |
| Present value over the working life (USD per girl moved out of NEET) | 4,843 |
| Girls exposed per girl moved out of NEET (1 / NEET reduction) | 14 |
| Cost per girl moved out of NEET (USD) | 2,710 |
| Return per USD 1, retention channel (central) | 1.79 |
| Skills premium of 5.6% for all exposed girls who work, if it held (USD per exposed girl, PV) | 365 |
| Return per USD 1 if the skills premium held | 3.61 |

*Data source: modelled projection (ex-ante); inputs as in key assumptions*

# Every USD 1 invested could return up to USD 3.61 in lifetime earnings




- **Both channels together: USD 3.61 per USD 1.** USD 723 in lifetime earnings per girl against a cost of USD 200.
- **Conservative view, staying in school only: USD 1.79 per USD 1.** USD 358 per girl, based on Indonesian causal evidence (PIP). The programme breaks even if it reduces girls' NEET by 4.1 points; the evidence suggests 7.4. At the 2025 minimum wage this rises to USD 2.48.
- **Skills channel on its own: USD 1.82 per USD 1.** USD 365 per girl if the 5.6% earnings premium for vocational schooling in Indonesia applies; that estimate is not statistically significant, so this is the less conservative part.
- **Not counted in either:** delayed marriage, health, children's outcomes and fiscal returns. These are benefit-cost ratios, not fiscal returns to government.
- **The USD 200 per girl** covers three years of digital content, teacher training, mentoring, shared connectivity and marriage-delay outreach, plus 15% overhead; PKH and PIP transfers are existing entitlements.

*Data source: modelled projection (ex-ante); inputs as in key assumptions*

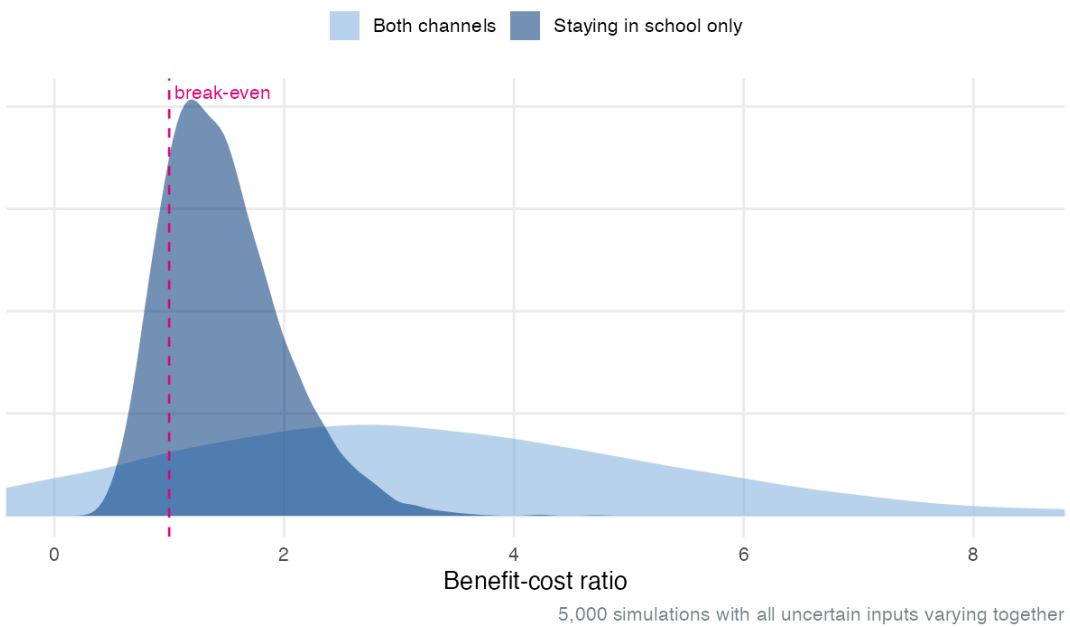
# Benefit-cost ratio by scenario

| Scenario | Retention benefit (USD/girl) | Skills premium benefit (USD/girl) | Total benefit (USD/girl) | Cost (USD/girl) | BCR |
|---|---|---|---|---|---|
| Central: retention channel (Indonesian causal evidence) | 358 | 0 | 358 | 200 | 1.79 |
| Targeted at Madura, retention channel | 615 | 0 | 615 | 200 | 3.07 |
| Targeted at PKH households, retention channel | 300 | 0 | 300 | 200 | 1.50 |
| Plus skills earnings premium of 5.6% (point estimate, not statistically significant) | 358 | 365 | 723 | 200 | 3.61 |
| Retention channel, earnings at the 2025 minimum wage | 496 | 0 | 496 | 200 | 2.48 |
| Retention channel, with Cash+ stipend for the poorest 40% | 358 | 0 | 358 | 359 | 1.00 |

*Data source: modelled projection (ex-ante); inputs as in key assumptions*

**Notes:** The central case is the retention channel. The targeted rows assume the programme's effect on school participation grows with the share of girls out of school in the group. In Madura more girls are out of school (30%, against 18% overall), so the same spending moves more of them. In PKH households fewer girls are out of school (15%), consistent with PKH's school-attendance condition, so a programme there adds less. The skills row adds the uncertain skills premium; the others show earnings at the 2025 minimum wage, and the cost of adding a Cash+ stipend for the poorest 40%. With other assumptions varied one at a time, the central return stays between USD 1.22 and USD 2.74 (appendix).

# How likely is the programme to pay for itself?




- **Counting only the benefit of staying in school**, the programme returns more than it costs in about 8 in 10 runs. A typical run returns USD 1.40 per dollar; most runs fall between USD 0.76 and USD 2.44.
- **Adding the skills effect**, it returns more than it costs in about 8 in 10 runs. The typical return is higher (USD 2.98), but the spread is much wider, because the evidence on the skills effect is weak; in some runs it is assumed to lower earnings, which is why the bottom of the range is below zero.
- The typical run (USD 1.40) is below our central estimate (USD 1.79) because some inputs, such as the discount rate and the employment rate, have more room to turn out worse than better.

*Data source: modelled projection (ex-ante); 5,000 simulations*

**Notes:** we re-ran the full cost-benefit calculation 5,000 times (a Monte Carlo simulation). In each run, every uncertain input took a different value within its plausible range: the programme's effect on school participation (from half to one and a half times the PIP estimate), the share of school leavers who become NEET, the return to schooling and the skills effect (each varying by its statistical margin of error), and earnings, employment, the discount rate and cost (within the ranges in the appendix). The chart shows how often each result came up.

# 4 · Key messages

# Key messages

**1.** Girls in East Java have the same school attendance rate as boys but are close to twice as likely to be NEET, a wider gap than nationally, costing about USD 4.9 billion in forgone lifetime productivity.

**2.** Completing senior secondary is associated with 20 points lower NEET for girls; child marriage with leaving school and more unpaid family work. NEET is highest in Madura, for girls and boys.

**3.** Most girls already stay in school (71% of girls aged 16–18 are in senior secondary), so a USD 200-per-girl programme reaching all girls changes the path of about 1 in 14 and returns USD 1.79 per dollar. Targeting Madura, where far more girls are out of school, raises this to USD 3.07. The next frontier is the transition into work: if skills training also raises earnings, the return could reach USD 3.61, an effect still to be tested in a pilot.

# Policy recommendations

**1. Target the transition, not enrolment.** Girls attend school at the same rate as boys, and senior-secondary enrolment rose faster in East Java than elsewhere in 2024–2025, but tertiary enrolment rose more slowly. Set targets on completion and the first job or further study, and pilot the skills component with an evaluation so its effect on earnings can be measured.

**2. Start in Madura.** NEET is highest and attendance lowest there: 30% of girls aged 16–18 are out of school, against 18% across East Java. Targeting Madura raises the projected return from USD 1.79 to USD 3.07 per dollar.

**3. Link girls to PKH and PIP rather than creating a new transfer.** Indonesian evidence shows PKH reduces girls' child marriage, and girls in PKH households are already less likely to be out of school (15%, against 18% overall). Linkage uses existing systems.

**4. Fund pedagogy and mentoring, not devices.** Internet use is already near universal, and devices alone did not raise learning in Uruguay.

**5. Measure the programme's own effect.** Programme districts improved more than the rest between 2024 and 2025 on every indicator, but not by enough to rule out chance. Build in a proper comparison from the start, and track whether girls move into wage employment rather than unpaid family work.

# Appendix

# Programme districts: comparison of changes

| Outcome | Programme 2024 | Programme 2025 | Other 2024 | Other 2025 | Difference in change (pp) | 95% CI | p |
|---|---|---|---|---|---|---|---|
| NEET | 26.9 | 22.8 | 25.9 | 23.3 | -1.5 | -5.7 to 2.7 | 0.487 |
| In school | 49.8 | 50.4 | 51.9 | 52.1 | 0.3 | -4.4 to 5.0 | 0.901 |
| Employed | 23.3 | 26.8 | 22.3 | 24.6 | 1.2 | -2.9 to 5.3 | 0.569 |
| Married before 18 | 6.2 | 4.3 | 4.9 | 4 | -1 | -3.1 to 1.0 | 0.331 |
| Senior-secondary enrolment (16-18) | 59 | 67.5 | 67.9 | 72.5 | 4 | -4.1 to 12.1 | 0.334 |
| Tertiary enrolment (18-20) | 21.5 | 26.7 | 21.2 | 23 | 3.3 | -4.4 to 11.1 | 0.396 |

**Notes:** Girls, East Java, survey-weighted (%). Difference in change: programme districts' change minus other districts' change, from a pooled survey-weighted regression with PSU-clustered standard errors.

# Regression results: child marriage and outcomes

| Hypothesis | Effect (pp) | SE | p | N |
|---|---|---|---|---|
| Child marriage → NEET | 28.7 | 4.6 | 0.0000 | 6060 |
| Child marriage → in school | -27.5 | 3.0 | 0.0000 | 6060 |
| Child marriage → employment | -1.2 | 4.6 | 0.7942 | 6060 |

**Notes:** Linear-probability models, girls 15–24, survey-weighted, PSU-clustered. **Controls: age and urban/rural.** Poverty is not controlled for because the 2025 expenditure module is not yet available. The association with employment is small and not statistically significant. Association, not causation.

# Child marriage and outcomes by region: regression results

| Region | Outcome | Effect (pp) | SE | p | N |
|---|---|---|---|---|---|
| Madura | Employed | 18.7 | 11.1 | 0.093 | 664 |
| Madura | NEET | 3.1 | 12.3 | 0.801 | 664 |
| Madura | In school | -21.8 | 9.8 | 0.026 | 664 |
| Rest of East Java | Employed | -4.6 | 5.1 | 0.361 | 5396 |
| Rest of East Java | NEET | 33.2 | 5.0 | 0.000 | 5396 |
| Rest of East Java | In school | -28.5 | 3.1 | 0.000 | 5396 |
| All East Java | Employed | -1.2 | 4.6 | 0.794 | 6060 |
| All East Java | NEET | 28.7 | 4.6 | 0.000 | 6060 |
| All East Java | In school | -27.5 | 3.0 | 0.000 | 6060 |

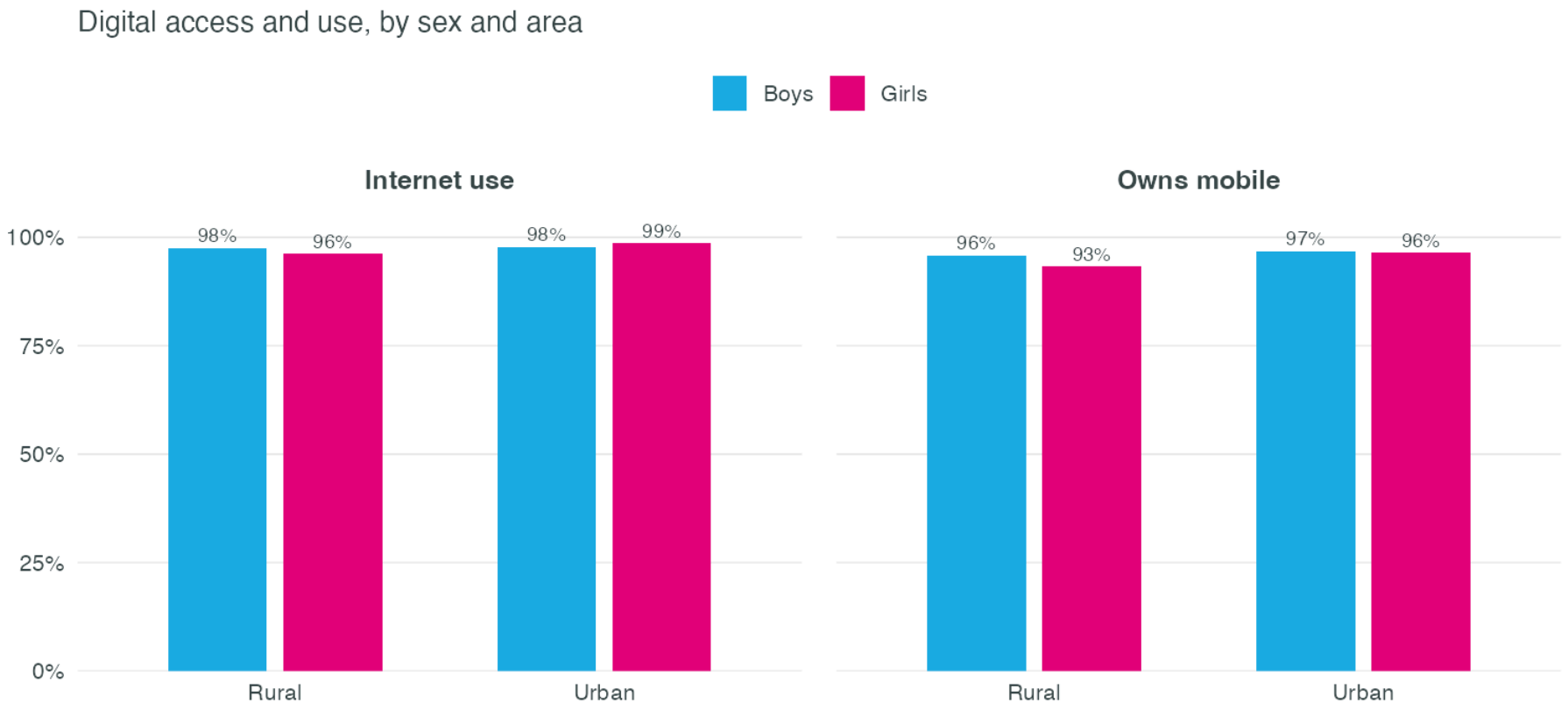
**Notes:** Linear-probability models, girls 15–24, survey-weighted, PSU-clustered; controls for age and urban/rural. Effects in percentage points.

# Earnings by type of work, young women (IFLS5)

| Type | Women 15-30 | With earnings reported | Median monthly earnings (IDR) |
|---|---|---|---|
| Employee (wage) | 1662 | 1652 | 1,100,000 |
| Own business | 689 | 667 | 500,000 |
| Casual worker | 203 | 203 | 250,000 |
| Unpaid family worker | 614 | 614 | 0 |

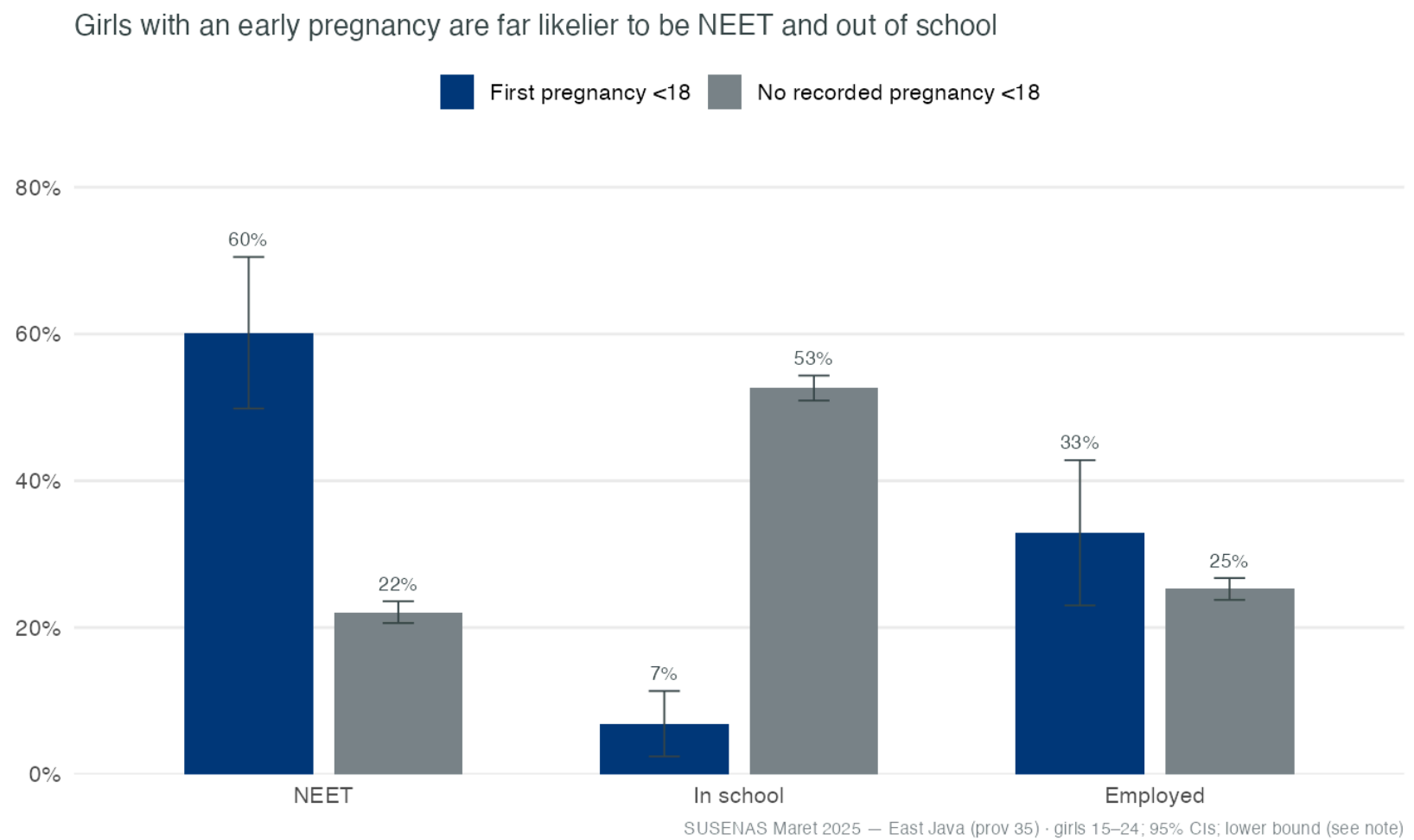
**Notes:** Women 15–30, IFLS5 (2014/15 prices). SUSENAS collects no earnings. Wages for employees and casual workers; net profit for own businesses; unpaid family workers have no earnings by definition.

# Digital access and use



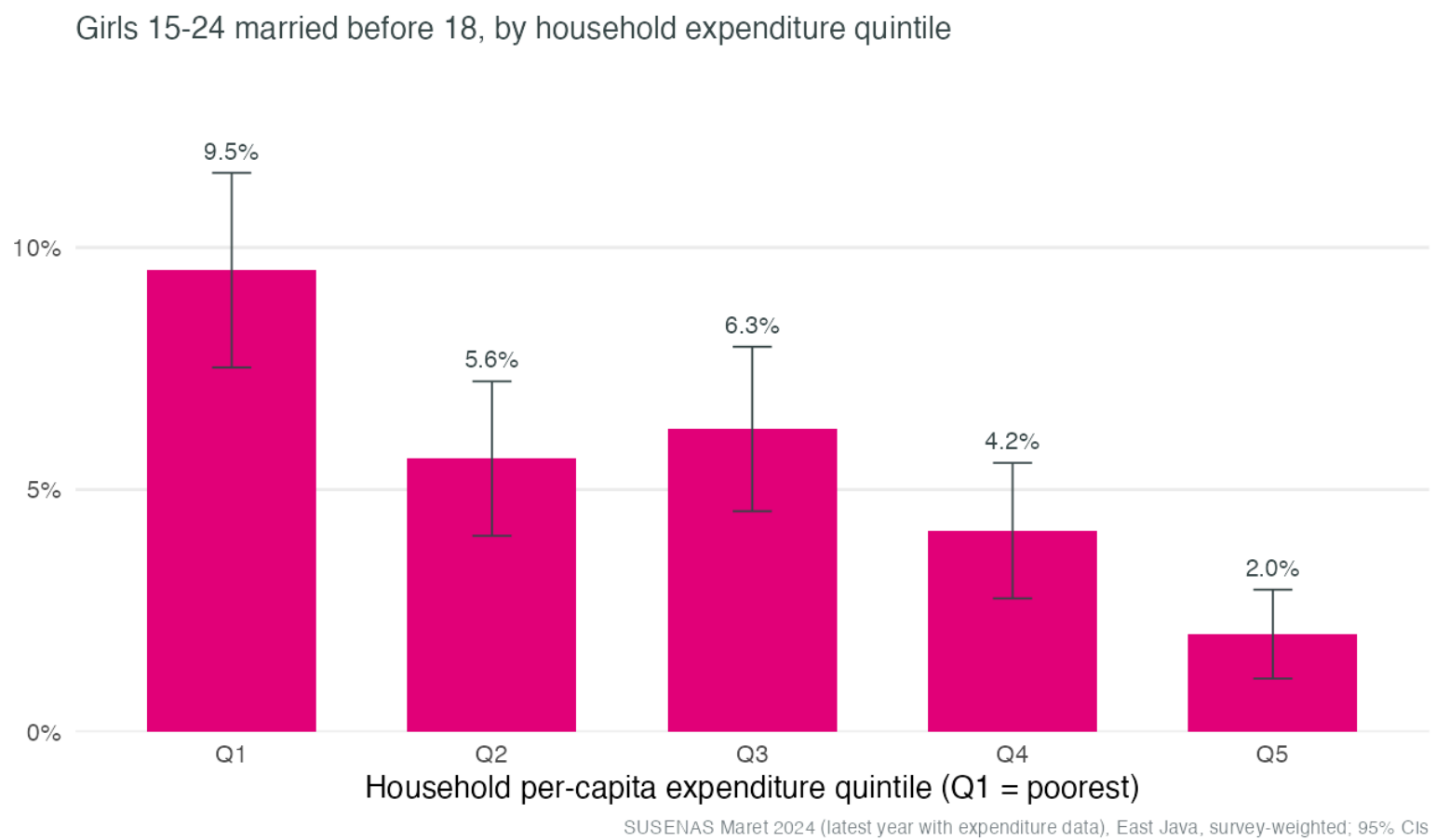

Source: SUSENAS Maret 2025 — East Java (prov 35). Weighted, adolescents 15–24.

# Early pregnancy and outcomes




**Notes:** Recorded only for ever-married women, so a lower bound. Girls with a first pregnancy before 18 are almost all girls married before 18, so the main deck reports child marriage only.

# Child marriage by household expenditure, 2024




**Notes:** SUSENAS 2024 is the latest year with household expenditure; the 2025 expenditure module has not yet been obtained.

# Child marriage and NEET by household PKH receipt

| region | PKH | n | Married_u18 | NEET |
|---|---|---|---|---|
| Madura | Not PKH | 509 | 6.1 | 35.9 |
| Madura | PKH household | 155 | 3.0 | 28.5 |
| Rest of East Java | Not PKH | 4661 | 4.1 | 22.1 |
| Rest of East Java | PKH household | 735 | 3.2 | 19.5 |

**Interpretation:** Girls 15–24, SUSENAS 2025. PKH targets poorer households, so lower rates among recipients are consistent with a protective effect, but are not a causal estimate.

# Cost components

| Component | USD/girl/yr | Source |
|---|---|---|
| Digital content & platform | 10 | Kemendikbudristek Merdeka Mengajar platform; INOVASI digital-learning unit costs |
| Teacher training (annualised) | 15 | Double Track PD days, annualised |
| Mentoring & socio-emotional | 10 | ELA/Skills4Girls facilitator costs |
| School connectivity/devices (shared) | 15 | Shared labs, no 1:1 devices |
| Ancillary: marriage-delay + flexible learning | 8 | ToR guidance: outreach to delay marriage, flexible options for caregivers |
| Annual direct cost per girl | 58 | Sum |
| Total per girl (3 years + 15% overhead) | 200 | Programme cost per girl |

**Notes:** These are benchmark unit costs from comparable programmes, not quotes from Double Track; they are placeholders until the provincial government shares actual budgets. The transfer itself is not costed because the Cash+ component links girls to existing PKH and PIP entitlements; a new stipend for the poorest 40% at the PIP senior-secondary rate would add about USD 46 per girl per year (ROI table).

# Sensitivity of the return to each assumption

| Parameter | BCR at low value | BCR at high value |
|---|---|---|
| Skills earnings premium (all exposed girls who work) | 1.79 | 3.61 |
| NEET reduction per exposed girl | 0.89 | 2.68 |
| Mincer return per year of edu | 1.26 | 2.74 |
| Discount rate | 1.22 | 2.20 |
| Annual cost per girl (sensitivity) | 2.30 | 1.38 |
| Female annual wage (East Java) | 1.34 | 2.23 |
| Real earnings growth | 1.33 | 2.10 |
| Female labour force participation | 1.29 | 1.86 |
| Years the skills premium lasts | 1.79 | 1.79 |

**Notes:** Each assumption varied on its own, others held at central values.

- **Low values**: a 2.5-point NEET effect (half the PIP-based estimate), the consumption-based return of 5.6% instead of 8%, earnings of Rp15m instead of Rp20m, a 5% discount rate, or a cost of USD 75 a year instead of 58.
- **High values**: a 7.5-point NEET effect, the 12.2% return without the selection correction, earnings of Rp25m, a 2% discount rate, or a cost of USD 45 a year.
- The effect size and the return to schooling matter most; no single assumption on its own takes the central case far from break-even.

# What the return counts

**Counted**

Higher schooling leads to a lifetime earnings premium for girls moved out of NEET, using the female return to schooling over a discounted working life.

**Excluded** (so the return is a lower bound)

Delayed child marriage · health and productivity spillovers · intergenerational gains · fiscal returns · broader empowerment

# Comparable international evidence

| Study | Design | Sample | Finding | Relevance for East Java |
|---|---|---|---|---|
| Bergstrom & Özler 2021 | Review of 104 interventions from 70 studies | Adolescent girls | Mixed evidence for life-skills training and girls' clubs; stronger for cash and schooling support | Supports cash and schooling components; cautions on clubs alone |
| Bandiera et al. 2018 (Sierra Leone) | RCT | Girls 12–25 | Enrolment +8.5pp during the Ebola period; effects on income-generating activity vary by age | Girls' clubs protected schooling in a crisis context |
| Aramburu, Goicoechea & Mobarak 2021 (Argentina, Colombia) | RCT | Women 18–50 | Coding bootcamps raised coding skills | Digital-skills content |
| Näslund-Hadley et al. 2025 (Paraguay) | RCT | Second-grade pupils | +0.09 SD computational thinking | Pedagogy; limited transfer to adolescent employment |
| Gülen et al. 2025 (Turkey) | Nonrandomized controlled pre/post | Girls 14–16 | Higher STEM aspirations and motivation | Girls' STEM engagement |
| Prakash et al. 2019, Samata (India) | RCT | Poor girls 13–16 | No overall effect on completion or marriage; schooling gains in one of two districts | Transition support alone had limited effects |
| Edmonds et al. 2021, Room to Read (India) | RCT | Girls | Dropout 4.1pp lower (main survey estimate) | Mentoring and life skills reduce dropout |
| Ashraf et al. 2025, Ten Years of Relational Power (Zambia) | RCT, 10-year follow-up | Girls ~14 | +0.23 years of schooling | Durable, low-cost soft-skills layer |
| de Melo et al. 2014, Plan Ceibal (Uruguay) | Quasi-experimental | Pupils | No detectable maths or reading gains in the first two years | Devices alone did not raise learning; fund pedagogy |
| Faizah, Isbah & Azca 2025 (online; 2026 issue) | Qualitative | Indonesian youth | Norms, care duties and marriage drive girls' NEET | Indonesia-specific NEET context |

**Notes:** Findings as reported in each source, with design and sample as verified. Used to inform programme design; the ROI effect comes from the Indonesian evidence. References on the last slide.

# Returns to schooling: specification and results

**Wage-based** (IFLS5 2014/15, adults 20–60, estimated separately for women and men, weighted):

$$\ln(w_i) = \beta_0 + \beta_1 S_i + \beta_2 \mathrm{Exp}_i + \beta_3 \mathrm{Exp}_i^2 + \beta_4 \mathrm{Urban}_i + \gamma_p + \phi_f + \phi_m + \delta\lambda_i + \varepsilon_i$$

$w$: monthly wage; $S$: years of schooling; Exp: potential experience (age minus schooling minus 6); $\gamma_p$: province fixed effects; $\phi_f$, $\phi_m$: father's and mother's education (fixed effects); $\lambda$: selection term from a first-stage model of working, in which children under 5 in the household predict whether a woman works but not her wage (Heckman two-step).

| Sex | Model | Return per year | SE | N |
|---|---|---|---|---|
| Women | (1) Baseline | 12.0% | 0.5 | 3830 |
| Women | (2) + Region | 12.2% | 1.1 | 3830 |
| Women | (3) + Family background | 10.9% | 1.1 | 3828 |
| Women | (4) + Selection (Heckman) | 8.0% | 1.2 | 3828 |
| Men | (1) Baseline | 9.4% | 0.3 | 6844 |
| Men | (2) + Region | 8.7% | 0.6 | 6843 |
| Men | (3) + Family background | 8.0% | 0.5 | 6842 |
| Men | (4) + Selection (Heckman) | 10.1% | 1.0 | 6842 |

**Notes:** The ROI uses the women's estimate with all controls (model 4, 8.0%). The consumption-based return from SUSENAS 2024 (5.6% for women; household per-capita expenditure regressed on schooling, schooling$^2$, age, age$^2$ and urban) is the low case in the sensitivity analysis.

# Data sources

- **SUSENAS Maret 2025** (BPS): NEET, school attendance and completion, child marriage, early pregnancy, type of work and hours, PKH receipt and digital access, for East Java and all provinces. East Java: 98,738 individuals, 12,599 aged 15–24, of whom 6,060 girls.
- **SUSENAS Maret 2024** (BPS): the 2024 comparison, and per-capita expenditure for the consumption-based return to schooling.
- **IFLS5 2014/15**: wages for the wage-based return to schooling.
- **Sakernas** (BPS): female labour force participation in East Java.
- **UMP Jawa Timur 2025**: benchmark for female earnings.
- **Published evaluations** of PKH, PIP and vocational training: the programme effect used in the return on investment.

All SUSENAS estimates are survey-weighted, with PSU/strata-clustered standard errors.

# Definitions

- **NEET (15–24)**: not in employment, education or training, built as not attending school (R611; 2024: R610) and not working in the last week or temporarily absent from a job (R703_A; R704, 2024: R705).
- **Married before 18**: age at first marriage (R409) under 18.
- **Senior-secondary completion**: highest certificate (R615) at SMA/SMK/MA level or above.
- **Type of work**: status in main job (R706); hours (R707).
- **Madura**: Bangkalan, Sampang, Pamekasan, Sumenep.

Several variables were renumbered between the 2024 and 2025 questionnaires (school attendance R610 to R611, temporarily absent R705 to R704, certificate R614 to R615, employment status R707 to R706); the activity block (R703) is the same in both years and both years are coded identically. The operational NEET definition does not separately exclude participants in non-formal training, which SUSENAS does not record.

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